Knowledge Graph - Lung Adenocarcinoma

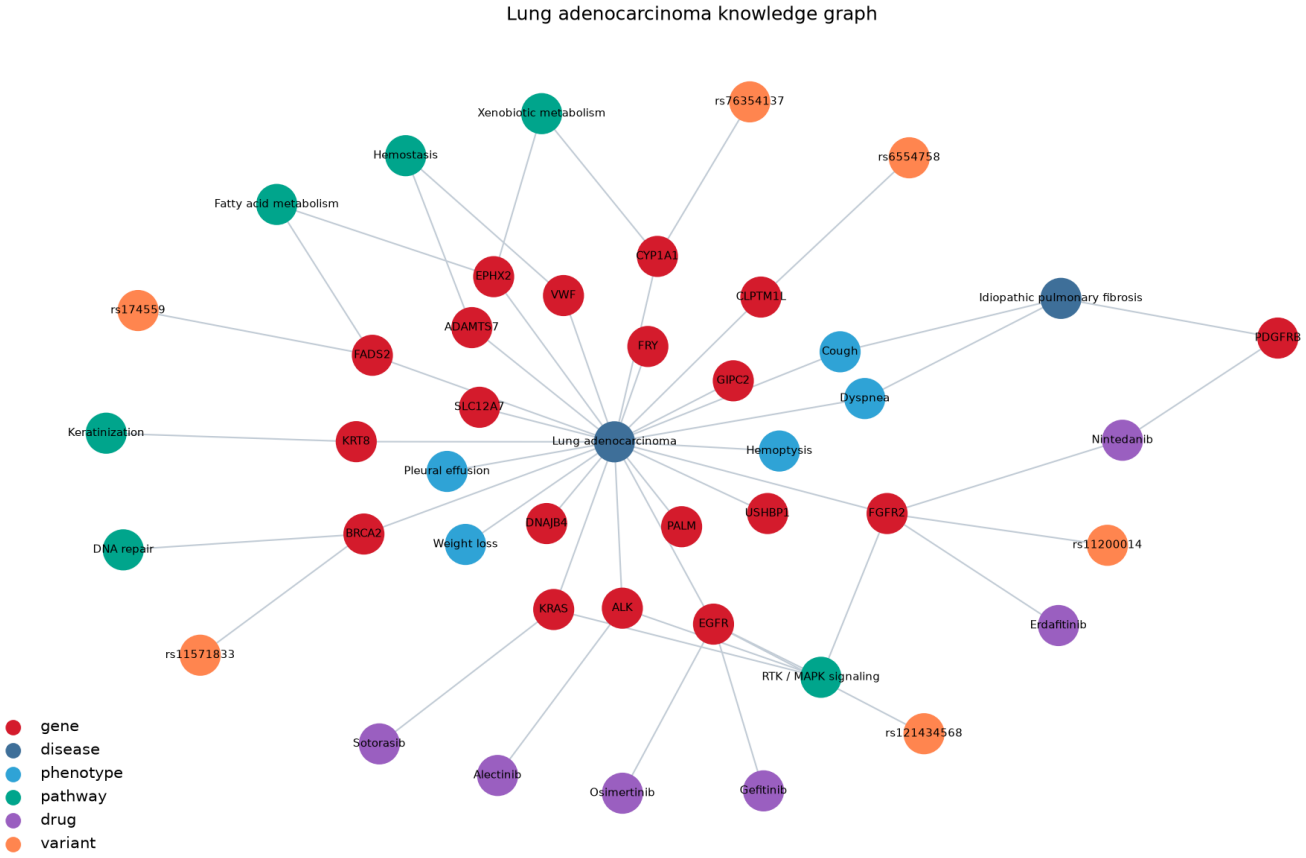
BIOT 6900 · Module 7 Assignment · Bhuvanesh Karthik Nilavalagan

Background

Earlier modules produced separate, unconnected outputs for lung adenocarcinoma: a ranked gene list, druggability scores, predicted structures and variant annotations. This assignment integrates them into a single queryable **knowledge graph of 44 nodes and 51 relationships**, **comprising 19 genes**, **6 pathways**, **6 drugs**, **6 variants**, **5 phenotypes and 2 diseases**.

1. Schema and evidence

Node types were selected based on what a drug discovery query must traverse. Gene nodes alone are insufficient: pathways group genes into mechanisms, drug nodes connect genes to therapeutics, variant nodes supply the evidence implicating each gene, and phenotype nodes capture the one attribute two otherwise unrelated diseases can share. The phenotype layer is what enables the repurposing query.



Edges are directional and type-specific rather than generic, because direction encodes meaning: a drug targeting a gene is a different assertion from a gene associated with a disease, and undirected links would permit biologically meaningless paths. Each edge carries source and confidence properties, arguably the most consequential design decision in the schema. GWAS edges derive from the GWAS Catalog with high confidence where the p-value exceeds genome-wide significance; drug edges from ChEMBL; pathway edges from Reactome. Edges added manually for the related disease carry medium confidence, having come from literature rather than a database. Without this labelling, all edges would appear equally reliable in query output, which they are not.

Identifiers were harmonized by keying each gene on its HGNC symbol with a single Ensembl ID via MyGene.info. One correction was required: CLPTM1L returned a scaffold identifier rather than the primary assembly, and was reassigned to ENSG00000049656. This is precisely the class of error that splits one gene into two nodes, distributing its centrality across both without any warning.

2. Top targets by centrality

PageRank scores

Gene	PageRank	Druggability
EGFR	0.0376	0.95
FGFR2	0.0331	0.86
FADS2	0.0196	0.55
ALK	0.0196	0.95
KRAS	0.0196	0.60
CLPTM1L	0.0196	0.39
BRCA2	0.0196	0.00
CYP1A1	0.0196	0.75

EGFR ranks highest, consistent with lung adenocarcinoma as the most established driver gene and therapeutic target in this disease. However, its position warrants qualification: it scores highly because two approved drugs target it and it sits in the RTK/MAPK pathway alongside three other genes, giving it the highest connection count. Those drug edges were added manually, so this partly reflects the constructed graph rather than an independent finding.

FGFR2 is the more robust result. It topped the Module 2 ranking on independent evidence and ranks second here for a different reason: a strong GWAS association, a variant, pathway membership and two drug edges all converge on it. Four independent categories of evidence are more persuasive than a high score from any single source.

The non-obvious hit is CYP1A1. It carries the weakest GWAS p-value among significant genes at 5e-07, yet ranks alongside BRCA2 and CLPTM1L. Its position derives from pathway membership rather than association strength, participating in two metabolic pathways with EPHX2: a different route to network importance than FGFR2 followed, and one visible only through the graph structure.

3. A hypothesis from a path

The repurposing query returned a single candidate, Nintedanib. The complete path is: Nintedanib, PDGFRB, idiopathic pulmonary fibrosis, cough, lung adenocarcinoma.

The hypothesis is that a drug developed for fibrotic lung disease may have activity in lung adenocarcinoma, since both involve growth factor receptor signaling. Nintedanib is a multi-kinase inhibitor that also inhibits FGFR family receptors, which is relevant because FGFR2 topped the Module 2 ranking. The graph consequently identified Nintedanib through two independent routes: directly as an FGFR2-targeting drug, and indirectly through the fibrosis path.

This would be tested using a cell proliferation assay using lung adenocarcinoma cell lines, including lines selected for FGFR2 alteration, treated with Nintedanib and measured against untreated controls. If the effect tracks with FGFR2 status rather than appearing uniformly, this supports a receptor signaling mechanism rather than general cytotoxicity. The path itself should not be over-interpreted, as cough and dyspnea are shared by most serious respiratory conditions; the mechanistic rationale provides the substantive support here, not the graph connection.

4. Reading one community

Community detection partitioned the graph into seven communities. The most informative contains CYP1A1, EPHX2, FADS2, xenobiotic metabolism, fatty acid metabolism, and the variants rs174559 and rs76354137.

This community represents carcinogen exposure and metabolism rather than growth signaling. CYP1A1 metabolically activates tobacco carcinogens, EPHX2 processes the resulting epoxide products, and FADS2 mediates fatty acid metabolism, with EPHX2 bridging both pathways. This is mechanistically distinct from the community containing EGFR, KRAS, ALK and their inhibitors, which concerns therapeutic targets in established tumours. The metabolic community instead relates to susceptibility and aetiology, with implications for prevention rather than treatment. The algorithm separated these two mechanisms without supervision.

5. An edge is evidence, not truth

Phenotype edges are on of the limitation. Cough, dyspnea and weight loss occur in nearly all serious respiratory conditions, so shared phenotype links carry minimal specificity. A more rigorous implementation would weight these edges by symptom rarity.

One limitation should be stated explicitly. A substantial block of genes ties at exactly 0.0196 or 0.0106. These ties reflect identical connection counts, not equivalent biological importance. PageRank measures network topology rather than evidence strength, and such ties are unavoidable at this graph size.

Drug-target edges are similarly simplified. Each drug was recorded as targeting a single gene, but Nintedanib inhibits multiple receptor families and Sotorasib is specific to one mutant form of KRAS. Collapsing these into single edges overstates the precision of the drug-gene relationships.

Use of AI was in three main areas. First, to write the Python that generates the dictionary for the nodes and edges from my disease data because the starter code provided loading and query but the data structure had to be created myself. Second, to find the Ensembl IDs corresponding to my fifteen gene symbols using MyGene.info instead of searching each of them separately. Finally, to analyze the results of the PageRank algorithm and community detection; in particular, to explain why all genes in a big chunk had the same score. I double-checked the output in two ways. One way was to calculate the total number of nodes again by hand to check if it equals 44. The second one was to verify all Ensembl IDs because they were responsible for the error in the CLPTM1L scaffold.

References

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